

Yajat Nagaraj Kiran

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EDUCATION

University of California, Irvine – Bren School of Information and Computer Sciences

Expected Graduation: June 2027

Bachelors of Science in Computer Science

- Awards: UC Davis Model UN Conference Best Research, FBLA Nationals 5th Place (Management Information Systems), California National Economics Challenge State Finalist
- Current coursework: Discrete Mathematics, Boolean Logic and Discrete Structures, Accelerated Programming in Python, Probability/Statistics for Computer Science, Software Engineering
- Community College Dual Enrollment Courses: Foundations of Data Science, Linear Algebra, Multivariable Calculus, International Business Management, Javascript, Machine Learning, Robotics and Embedded Systems

PROFESSIONAL EXPERIENCE

Wehab Lab @ Stanford University

June 2024 - August 2024

Research Assistant

Stanford, California

- Designed and prototyped a haptic breathing pacer app using Figma for children with autism, increasing pilot engagement scores by 32% in initial usability tests
- Synthesized user insights from 150+ survey responses to inform product roadmap decisions and feature prioritization with Qualtrics
- Engineered and deployed a new lab website with JavaScript and Gatsby

Neptune Research Group

December 2023 - August 2025

Researcher

San Jose, California

- Engineered deep learning pipelines in TensorFlow and Keras to classify abnormal brain activity from spectrograms, improving detection accuracy by 18%
- Leveraged transfer learning + wavelet transforms for anomaly detection, reducing feature-processing time by 40%
- Achieved a KL divergence score of 0.43; results published in IEEE MIPR 2024 under “*Harmful Brain Activity Classification of Spectrograms with Transfer Deep Learning*”

UC Berkeley

August 2024 - June 2025

Researcher under Dr. Jeff Perloff

San Jose, California

- Applied Graph Neural Networks (GNNs) for blockchain fraud detection, boosting classification performance by 12% ROC-AUC
- Built preprocessing pipelines with NumPy/Pandas/Scikit-learn (normalization, PCA) to reduce feature dimensionality
- Implemented ensemble models (RF + Gradient Boosting), improving precision-recall AUC by 20% in imbalanced fraud detection tasks
- Authored and presented research paper “*Enhancing Elliptic Curve Cryptography with Graph Neural Networks and Ensemble Methods: A Hybrid Approach*” at IEEE ICIIS 2024

UC Irvine ZotBins

September 2025 - present

Computer Vision Researcher

Irvine, California

- Designed cloud-based waste recognition system on AWS free tier utilizing EC2, RDS PostgreSQL, and S3 for processing real-time sensor data from ESP32 hardware for smart waste management
- Wrote FastAPI REST API scripts with operations for inference management in order to process 1000+ daily predictions
- Built processing pipeline using OpenCV and NumPy to decode images, run batch inference, and generate masked crops with class labels

LEADERSHIP & EXTRACURRICULARS

Nysha.ai

August 2022 - June 2025

Vice President

Fremont, California

- Directed AI/ML projects (one accepted to IEEE conference), integrating machine learning models for cystic fibrosis research
- Implemented cloud-based volunteer tracking using HubSpot, onboarding and managing 200+ global volunteers
- Expanded organizational presence to 15 states and 5 countries, raising \$27,000 for cystic fibrosis research

SKILLS

Languages: Python (NumPy, Pandas, Scikit-learn, PySpark)

Frameworks / Libraries: TensorFlow, Keras, Pytorch, OpenCV, PyWavelets, Gatsby, FastAPI

Tools & Platforms: AWS, Git/Github, VS Code, Figma, Jupyter Lab, Qualtrics, Photoshop, After Effects